

Search for events in XENON1T associated with Gravitational Waves

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(XENON Collaboration)

We perform a blind search for events in the XENON1T dark matter detector that occur close in time to gravitational wave signals in the LIGO and Virgo observatories. No events are observed in the nuclear recoil, electronic recoil, $\text{CE}\nu\text{NS}$, and S2-only channels within ± 500 seconds of observations of the gravitational wave signals GW170104, GW170729, GW170817, GW170818, and GW170823. We use this null result to constrain mono-energetic neutrinos and Beyond Standard Model particles emitted in GW170817, a binary neutron star merger. We set new upper limits on the fluence of coincident neutrinos down to 16 keV at 90% confidence level. Furthermore, we constrain the product of coincident fluence and cross section of Beyond Standard Model particles to be less than 10^{-29} in the [5.5–210] keV energy range at 90% confidence level.

I. INTRODUCTION

II. DATA SETS

Following the first detection of a gravitational wave (GW) signal by the Laser Interferometer Gravitational-Wave Observatory (LIGO) [1], signals from binary mergers are now routinely detected by the LIGO, Virgo, KAGRA and GEO observatories [2]. The subsequent observation of electromagnetic counterparts from binary neutron star coalescence [3] has made GW events an integral part of multi-messenger astrophysics. Various detectors have conducted follow-up searches for neutrinos associated with binary mergers in a time window of $\pm [400\text{--}500]$ s of the observed GW signal [4–10], which might indicate the simultaneous emission of relativistic particles such as neutrinos. The sensitive (neutrino) energy ranges of these searches, summarized in Tab. I, span from EeV down to half an MeV. No evidence of a coincident signal has been reported yet. Here, we use data from the XENON1T dark matter search [11] to extend this neutrino energy window down to 16 keV. Further, our low threshold of 1 keV electron-equivalent (keV_{ee}) enables us to search for Beyond Standard Model (BSM) particles from the merger of neutron stars, such as relativistic axions [12, 13] or sterile neutrinos [14, 15].

The LIGO and Virgo observatories conducted their second observation run from November 30, 2016, to August 25, 2017 [21]. Overlapping with this time frame, the XENON1T detector took Science Run 0 (SR0) data from November 22, 2016 to January 18, 2017, and Science Run 1 (SR1) data from February 2, 2017 to February 8, 2018. This data has already been analyzed in several channels that we revisit here in order to check for events coincident with LIGO and Virgo GW observations. A search for nuclear recoils (NR) demanding both scintillation light (S1) and electroluminescence from ionization electrons (S2) is available during both SR0 [16] and SR1 [17]. In addition, using data from SR1, a search was conducted for electronic recoils (ER) [18], for coherent elastic neutrino-nucleus scattering ($\text{CE}\nu\text{NS}$) [19], and for nuclear recoils where the requirement of a scintillation signal has been waived (S2-only) [20].

Details of these analysis channels can be found in the respective papers [16–20]. Notably, the NR and ER channels demand coincident signals from at least three photomultiplier tubes (PMTs) for an S1, while the $\text{CE}\nu\text{NS}$ channel requires signals from only two PMTs. Furthermore, the S2 threshold has been reduced from 200 photoelectrons (PE) and 500 PE for the NR and ER channels, respectively, to 120 PE and 150 PE for the $\text{CE}\nu\text{NS}$ and S2-only channels, respectively. The corresponding energy ranges of these channels are given in Tab. II in units of keV electron-equivalent (keV_{ee}) for the ER channel, and as keV nuclear recoil-equivalent (keV_{nr}) for the NR, $\text{CE}\nu\text{NS}$, and S2-only channels. Effective xenon target masses vary for different analyses. Whereas the fiducial volume limits the exposure of the NR, ER, and $\text{CE}\nu\text{NS}$ channels, the S2-only channel is limited by the strong event selections applied to reduce backgrounds.

Background rates in the NR, $\text{CE}\nu\text{NS}$, and S2-only channels are only a few events (14, 6, and 170, respectively) observed during the entire Science Runs. This motivates the use of a single energy bin for each of these channels. From Fig. 3 of [18], we can see that the ER spectrum is not constant in energy. The spectrum ex-

TABLE I. Previous searches for events coincident with GW and the neutrino energy ranges they are sensitive to.

Detector	Detection energy
Pierre Auger Observatory [4]	100 PeV – 25 EeV
IceCube [5]	100 GeV – 100 PeV
Super-K [6]	3.5 MeV – 100 PeV
XMASS [7]	14 MeV – 100 MeV
KamLAND [8]	1.8 MeV – 111 MeV
NOvA [9]	1 MeV – 10 MeV
Borexino [10]	0.5 MeV – 5 MeV

TABLE II. Characteristics of analysis channels considered in this study. For each of these channels, we show the corresponding energy ranges, the Science Runs that have been analyzed for that channel, the fiducial target mass, and the observed event rate. Further, in the ± 500 s time window around a GW event, we give the average livetime for detecting a signal, the average number of expected events (N_{exp}), and the probability of observing zero events ($P(0)$).

Event channel	Energy range	Runs analyzed	Fiducial target mass	Event rate (events/day)	Livetime (per GW)	N_{exp}	$P(0)$
NR [16, 17]	4.9 keV _{nr} – 40.9 keV _{nr}	SR0, SR1	1.3 tonne	0.05	935 s	0.00054	0.9995
ER [18]	1 keV _{ee} – 210 keV _{ee}	SR1	1.042 tonne	186	935 s	See Tab. IV	See Tab. IV
CE ν NS [19]	≥ 1.6 keV _{nr}	SR1	1.04 tonne	0.03	790 s	0.00027	0.9997
S2-only [20]	≥ 0.7 keV _{nr}	SR1	0.121 tonne	0.94	650 s	0.0071	0.993

hibits spectral lines from $^{83\text{m}}\text{Kr}$ and $^{131\text{m}}\text{Xe}$. We thus define five energy bins for the ER channel, see Tab. IV: low energy (0–30 keV_{ee}), $^{83\text{m}}\text{Kr}$ (30–50 keV_{ee}), medium energy (50–142 keV_{ee}), $^{131\text{m}}\text{Xe}$ (142–185 keV_{ee}) and high energy (185–210 keV_{ee}).

Three of the eight GW events that LIGO/VIRGO observed during SR0 and SR1 occurred during XENON1T calibration or downtime. Only SR0 NR data is available during GW170104, while SR1 data in all four analysis channels is available during GW170729, GW170817, GW170818, and GW170823. Out of these, GW170817 was due to the merger of neutron stars. Since this merger was at a distance of 40 Mpc, compared to a distance of ~ 1000 Mpc for other events, particle flux limits from that event are dominant.

III. SIGNAL SPECTRA

Neutrinos predominantly interact in our detector via elastic scattering off electrons or xenon nuclei. The standard model differential cross section for elastic neutrino-electron scattering can be written as [22, 23]:

$$\frac{d\sigma_{\text{ES}}}{dE_r} = \frac{G_F^2 m_e}{2\pi} \left[(g_V + g_A)^2 - (g_V^2 - g_A^2) \frac{m_e E_r}{E_\nu^2} + (g_V - g_A)^2 \left(1 - \frac{E_r}{E_\nu} \right)^2 \right], \quad (1)$$

where E_r is the electron recoil kinetic energy, E_ν is the incoming neutrino energy, m_e is the mass of an electron, and G_F is the Fermi constant. Neutrino-electron coupling constants for different flavors of neutrinos and antineutrinos are furnished in Tab. III.

The expected number of signals in the ER channel is given as:

$$\frac{dN_{\text{ER}}}{dE_r} = N_T(E_r) \epsilon(E_r) \int \frac{d\sigma_{\text{ES}}}{dE_r} \frac{df(E_\nu)}{dE_\nu} dE_\nu, \quad (2)$$

where $\epsilon(E_r)$ is the efficiency of ER channel as a function of the recoil energy [18]. $N_T(E_r)$ is the total number of available electrons in the detector, which is obtained by multiplying the total number of xenon atoms in the fiducial volume with the number of electrons per atom with

TABLE III. Neutrino-electron coupling constants for different flavors of neutrinos and antineutrinos. θ_W is the Weinberg angle.

Neutrino flavor	g_V	g_A
ν_e	$2 \sin^2 \theta_W + \frac{1}{2}$	$+\frac{1}{2}$
$\bar{\nu}_e$	$2 \sin^2 \theta_W + \frac{1}{2}$	$-\frac{1}{2}$
ν_μ, ν_τ	$2 \sin^2 \theta_W - \frac{1}{2}$	$-\frac{1}{2}$
$\bar{\nu}_\mu, \bar{\nu}_\tau$	$2 \sin^2 \theta_W - \frac{1}{2}$	$+\frac{1}{2}$

binding energies above the particular recoil energy. The different shells of a xenon atom, along with the number of electrons and the potential energy, are given in [24]. It should be noted that the lower energy threshold for ER channel is 1 keV_{ee} [18].

The effective cross section as a function of neutrino energy, taking into account the detector properties, is shown in Fig. 1. We convolve the cross section with a Gaussian function with standard deviation $0.31\sqrt{E} + 0.0037E$ [18] to account for energy resolution. We also account for the fact that a neutrino can not produce a recoil of energy greater than $2E_\nu^2/(2E_\nu + m_e)$ [23]. Our threshold of 1 keV_{ee} makes us insensitive to neutrinos below 16 keV. Given an absence of theoretical predictions, we assume that neutrinos are emitted mono-energetically, as done in previous low-energy searches [6, 7, 10]. The cross section for electronic recoils depends weakly on neutrino energy, making it almost constant at neutrino energies above 36 keV. However, it decreases below 36 keV because low-energy neutrinos are incapable of generating electronic recoils throughout the entire energy range of ER channel, as shown in Tab. II.

Having discussed the interaction of neutrinos with electrons, we now turn to the interaction of neutrinos with xenon nuclei. For coherent elastic neutrino-nucleus scattering, we use the same differential cross section as in [19]. The number of nuclear recoils in the detector has the same form as Eq. (2). Assuming mono-energetic emission of the neutrinos, the effective cross section as a function of neutrino energy, taking into account the detector properties, is shown in Fig. 1. The values of non-standard

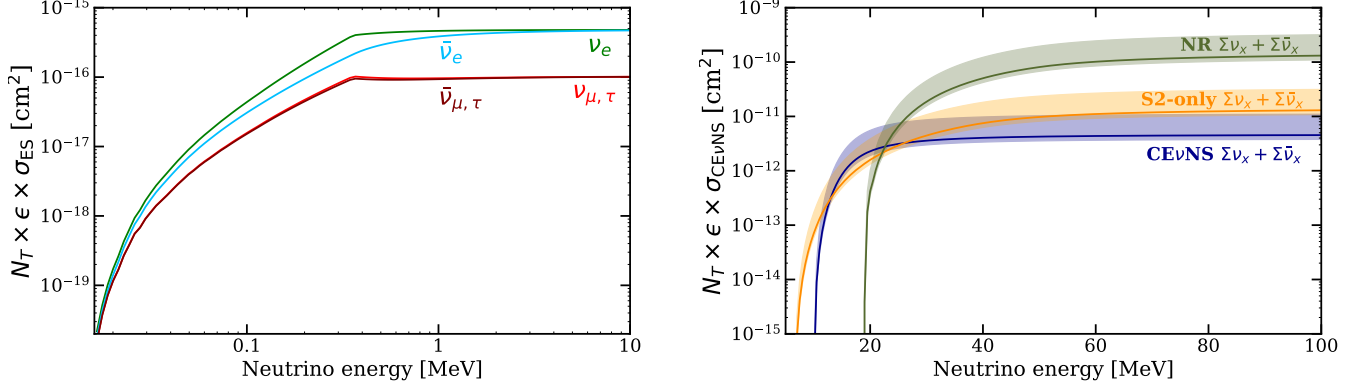


FIG. 1. Left: the effective cross section for elastic neutrino-electron scattering as a function of neutrino energy for electron neutrinos (ν_e , green), electron antineutrinos ($\bar{\nu}_e$, sky blue), muon or tau neutrinos ($\nu_{\mu,\tau}$, red), and muon or tau antineutrinos ($\bar{\nu}_{\mu,\tau}$, maroon). Right: the effective cross section for CE ν NS as a function of neutrino energy. NR (olive-green), CE ν NS (blue), and S2-only (orange) refer to different ways in which the data was analyzed in the search for nuclear recoils, as described in Sec. II. In the right plot, the cross section is for the interaction of all flavors of neutrinos and antineutrinos.

couplings ϵ_{ee}^{dV} and ϵ_{ee}^{dV} permitted by XENON1T, allows the effective change in the total interaction cross section from a factor of 0.83 to 2.42, under the assumption that $1/6^{th}$ of the total neutrinos are electron neutrinos. The resulting changes in the number of nuclear recoils are shown as an uncertainty band.

IV. BACKGROUND ESTIMATION

Consistent with previous coincidence searches shown in Tab. I, which used a time window of $\pm [400-500]$ s, we use a ± 500 s time window centered on the GW event time. This is appropriate for any relativistic particle flux emitted by the binary merger. Accounting for deadtime due to various cuts and background events, livetimes for detecting a signal in the ± 500 s time window around a GW event in the NR, ER, CE ν NS and S2-only channels are approximately 935 s, 935 s, 790 s and 650 s respectively. Fig. 2 shows the distribution of background events with time in different analysis channels. Rates are constant within statistical uncertainty for the duration of the Science Runs for the NR, CE ν NS, and S2-only channels. Therefore, for these channels,

$$\text{event rate} = \frac{\text{total events during entire Science Run}}{\text{total livetime of entire Science Run}}. \quad (3)$$

However, due to the background of activated isotopes — ^{131m}Xe , ^{133}Xe , and ^{125}I ; this is a poor approximation for the ER channel. Thus, we use the same background estimation for the ER channel as in [25]. The rate of ER events, as predicted by the background models in different energy bins, is shown in Fig. 3. Before unblinding the event timing information, we compared the values predicted by these background models with the observed values for all ER channels using the $(-1500, -500)$ s side-

band before the actual GW events. Given five independent ER energy bins and four GW events, we divided the chi-square by 20 to find the reduced chi-square statistic to be 1.14, confirming that these background models agree with the observation.

In the ± 500 s time window around a GW event, the expected number of background events is calculated as

$$N_{\text{exp}} = \text{event rate} * \frac{\text{lifetime}}{24 * 60 * 60}. \quad (4)$$

The corresponding probabilities of observing zero events are values of cumulative distribution functions at zero, assuming Poisson statistics. These values are given in Tab. II for the NR, CE ν NS, and S2-only channels and in Tab. IV for the ER channel.

While the ± 500 s time window was still blinded during background estimation, we calculated an upper limit on the number of expected events in the ER channel, based on the background model, above which evidence for a signal could have been claimed. We used Poisson statistics with a trial factor t_f of 20 and applied the Sidak Correction [26]:

$$p_{\text{global}} = 1 - (1 - p_{\text{local}})^{1/t_f}. \quad (5)$$

The local p-value of 0.003 corresponds to a global p-value of 0.00015 for a potential excess at a 3σ confidence level. The resulting upper limits on the expected background events are between 2 and 5 events, as shown in Tab. IV. We do not give upper limits on the number of expected events in other channels because values of cumulative distribution functions at 1, for Poisson distributions with such small means, are more than 99.99%.

TABLE IV. Predictions for the different energy bins of the ER channel during four GW events. For each bin, in the ± 500 s time window around a GW event, we give the number of expected events (N_{exp}), the probability of observing zero events ($P(0)$) and the upper limits (UL) on the expected background events at a 3σ confidence level.

	Low Energy			$^{83\text{m}}\text{Kr}$			Medium Energy			$^{131\text{m}}\text{Xe}$			High Energy		
	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL
GW170729	0.04	0.94	2	0.16	0.85	3	0.16	0.80	3	0.64	0.52	5	0.07	0.93	2
GW170817	0.04	0.94	2	0.15	0.86	3	0.15	0.80	3	0.29	0.75	4	0.07	0.93	2
GW170818	0.04	0.94	2	0.15	0.86	3	0.15	0.80	3	0.27	0.76	4	0.07	0.93	2
GW170823	0.04	0.94	2	0.14	0.87	3	0.15	0.80	3	0.23	0.79	3	0.07	0.93	2

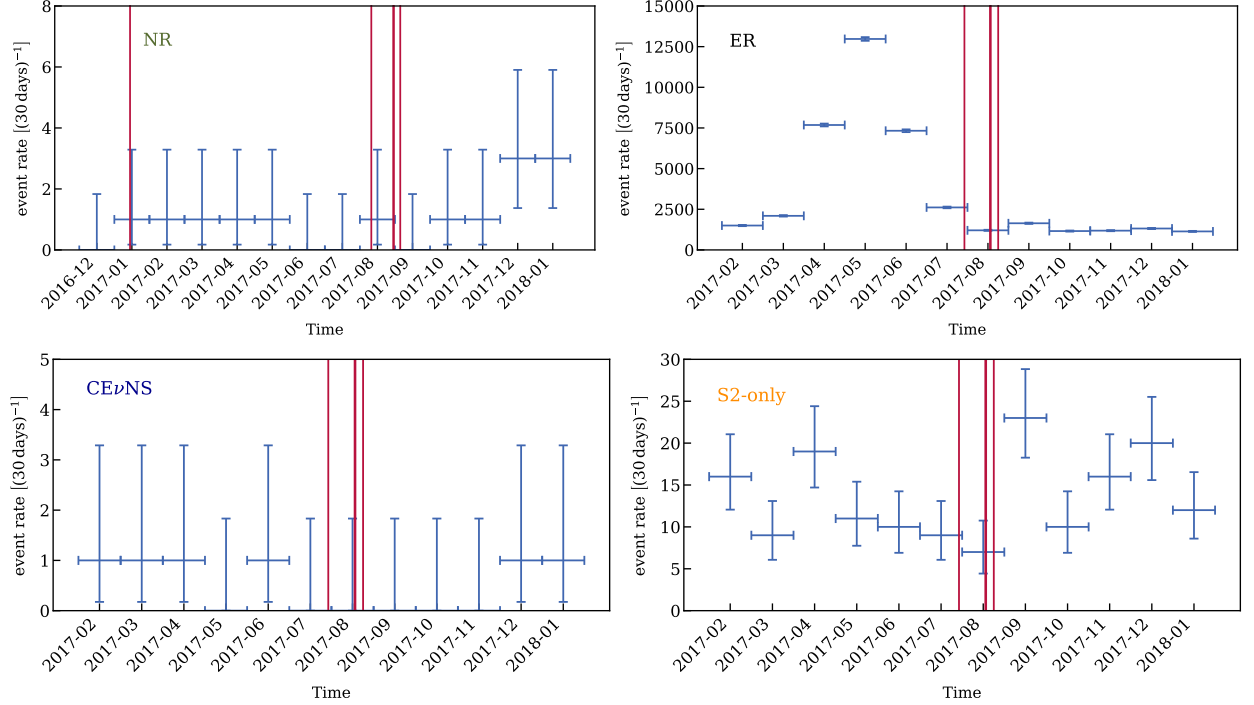


FIG. 2. We show the distribution of background events with time in different analysis channels. Red lines indicate the time of GW events. The rise and subsequent decay of event rate in the ER channel during March-September are due to production of $^{131\text{m}}\text{Xe}$, ^{133}Xe , and ^{125}I during neutron calibration [25].

V. RESULTS AND CONCLUSIONS

Once these detection channels, time and energy ranges, and background models were fixed, event timing information was unblinded. No event was observed in any of these channels within ± 500 seconds of any tested GW event. Given the low background rates, this is expected under the background-only hypothesis in the NR, CE ν NS, and S2-only channels. The total number of expected background events in the ER channel across all four GW events is ~ 3 . However, even in this channel, no events were observed, consistent with the background-only hypothesis, assuming Poisson statistics, within a 3σ confidence interval.

The absence of events can be converted into an upper limit at 90% confidence level on the fluence of coincident neutrinos emitted in the merger of neutron stars –

GW170817, assuming a Poisson distribution of expected events [27]. The upper limit on neutrino fluence at 90% confidence level using electronic recoils is shown in Fig. 4. As can be seen, we place upper limits on the fluence of $\mathcal{O}(10 \text{ keV})$ neutrinos. No other detector has been able to probe such low-energy neutrinos. The upper limit on neutrino fluence at 90% confidence level using nuclear recoils is shown in Fig. 5. As can be seen, our limits are comparable to those from the XMASS detector. It should be noted that our limits, as well as those from the XMASS detector, are on the sum of the fluence of all flavors of neutrinos and antineutrinos.

We also use our data to constrain the fluence of BSM particles with keV energies coincident with GW events. Similar to neutrinos, we assume that these BSM particles are emitted mono-energetically. Further, in the absence of a well-defined differential cross section for the inter-

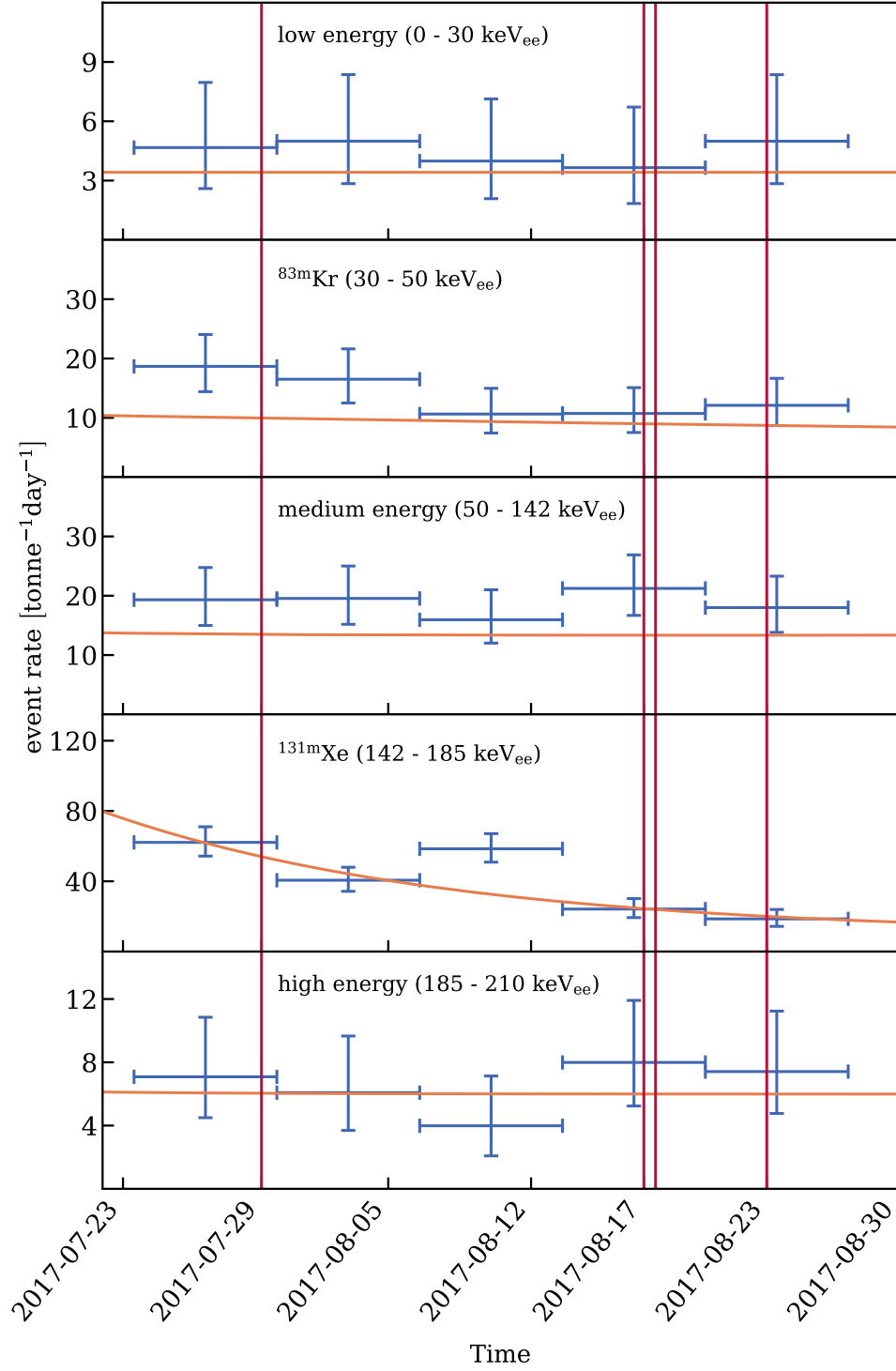


FIG. 3. We show the rate of ER events in different energy bins. Red lines indicate the time of GW events. Orange lines indicate the rate predicted by the background models as described in [18].

action of these BSM particles with electrons, we assume that all of their energy is transferred to the electron during an interaction. For this assumption to be valid, the particles' rest mass must be negligible compared to their total energy of $\mathcal{O}(10 \text{ keV})$. These assumptions yield the number of electronic recoils in the detector due to BSM particles as:

$$N_{\text{ER}} = N_T \epsilon \sigma_{\text{BSM}} F_{E_{\text{BSM}}}, \quad (6)$$

where σ_{BSM} is the cross section of BSM particles to inter-

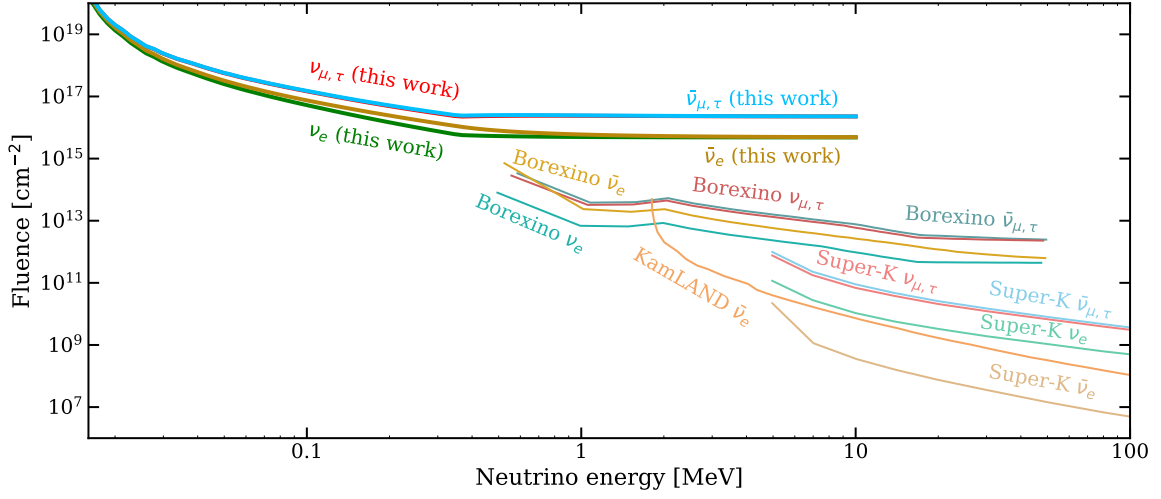


FIG. 4. The upper limit on the fluence of coincident neutrinos emitted in the merger of neutron stars using the electronic recoil signals in XENON1T, obtained for electron neutrinos (ν_e , green), electron antineutrinos ($\bar{\nu}_e$, sky blue), muon or tau neutrinos ($\nu_{\mu,\tau}$, red), and muon or tau antineutrinos ($\bar{\nu}_{\mu,\tau}$, maroon). We also show the limits obtained by Borexino [10], KamLAND [8], and Super-K [6] for comparison.

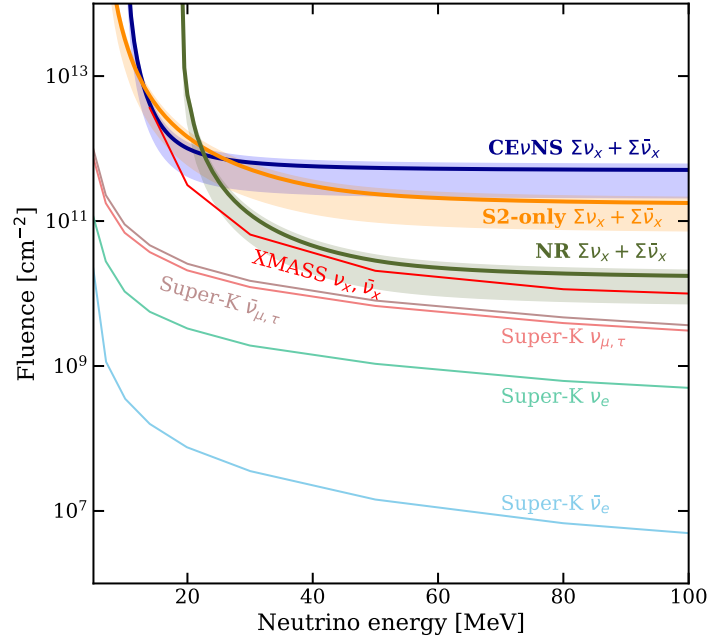


FIG. 5. The upper limit on the fluence of coincident neutrinos emitted in the merger of neutron stars using the nuclear recoil signals in XENON1T, placed on the sum of the fluence of all flavors of neutrinos and antineutrinos. NR (olive-green), CEνNS (blue), and S2-only (orange) refer to different ways in which the data was analyzed in search for nuclear recoils, as described in Sec. II. We also show the limits obtained by XMASS [7] and Super-K [6] for comparison.

act with electrons and $F_{E_{\text{BSM}}}$ is the fluence of BSM particles with energy E_{BSM} . It can be seen in Fig. 2 of [18] that the efficiency of our detector for the ER channel is $\sim 89\%$ for recoil energies greater than 5 keV_{ee}. The number of electrons available per atom when the recoil energy is greater than 34.6 (5.5) keV_{ee} is 54 (52) electrons. Given that we do not see any event, we obtain the upper limit on the product of coincident fluence and cross section: $\sigma_{\text{BSM}} F_{E_{\text{BSM}}} < 10^{-29}$ for E_{BSM} in [5.5–210] keV_{ee}.

We have thus provided the first search for signals in the keV energy region associated with GW events. Despite the 1.3 tonnes fiducial mass of XENON1T, energy regions going down to 1 keV_{ee}, and multiple nuclear and electronic recoil channels, no signals were found within ± 500 s time windows around GW events. That enabled

us to extend the upper limits on the fluence of coincident neutrinos down to 16 keV. Further, we use the lack of signals to put an upper limit on the product of coincident fluence and cross section of BSM particles. These limits can help constrain models involving neutrinos or other BSM particles in binary star mergers. As gravitational wave detectors will continue taking data during the operation of the current and next generation of xenon detectors [28–32], future detections of nearby mergers will further improve these constraints.

VI. ACKNOWLEDGEMENTS

We are grateful...

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